

ActInf GuestStream #006.1 -- Anna Ciaunica

Source: [YouTube](#) Playlist: GuestStreams Duration: 1:22:49 | Uploaded:
Unknown Transcript method: auto_caption

hello let me just wait one sorry just one second so you introduced me first right yep just wanted to make sure that it started and it looks like we have so hello and welcome everyone to active lab this is actin flap guest stream number 6.1 and we're really happy to be here with dr anna cia so anna thanks so much for joining we really appreciate it and we're looking forward to this talk please just take it away and if anyone in the uh live chat wants to ask a question i will relay those questions to you at the end of this talk so thanks again for coming on the stream and we're looking forward to what you have to share thank you so much danielle and thank you so much for having me um today i'm going to present a paper that we worked on the past past year and just came out um but i'm going to start very briefly with some of my background as a researcher so i'm trained as an armchair philosopher so i did um in analytics philosophy so i did my thesis on physicalism in qualia you know the famous mind body problem so how exactly so some physical stuff connects with you know mental stuff and gets you know the uh gives rise to the subjective experience of our conscious experiences like the heart problem of consciousness um and i was obsessed as any respectable philosopher i reality philosophy with you know the whole problem of consciousness and whether there is something special about the subjectivity of our conscious experiences right and because i'm coming from burgundy i spent a lot of time trying to you know say that there is something about our there is something special about our subjective conscious experience because there is something what is this like for instance to taste a particular wine and coming from burgundy i had a lot of experience and natural example with that but more recently i decided to get off the armchair of philosophy and i'm recently started to collaborate with people from quantum science i actually also have a master in cognitive science in development of psychology and uh i'm recently collaborated with the institute of community of neuroscience as ucl in london and my job there is to ask annoying questions to uh scientists like what do you mean by perception what do you mean by social right um just to give us some examples so uh one of the annoying questions i really i like to ask is um you know whenever i go to a conference is uh i ask people to look around them in the audience and realize that actually there are no

infants in the audience uh just adults and uh i interpolate them on that and they look at me say you know why i don't see the point why is this important well it is important because i mean when we start investigating mind and bodies we already have another centric um theoretical lens so this means that the way we look at the phenomena is from a certain standpoint and the standard point that you're using inverted um inescapably is an adult one right so this means that we might have a sort of like implicit bias towards certain phenomena a certain way of understanding phenomena that is somehow you know um adult centric and uh we need to be careful if we want to understand her mind and uh how mind and brain work in the first place so um given that my initial uh you know entrance point into the science and philosophy was like uh perceptual awareness like the fundamental point of contact of a subject with the reality of the objects out there in the world so um this is like the fundamental way of actually structuring an experience so you have this axis you have an experience of subject with relations object over there and typically for instance this is how self-consciousness is understood like uh perceptual awareness when one perceives oneself as self-distinct from the world and others and how is that linked to self-recognition in the mirror for instance right so for instance like is the famous rouge test if you recognize yourself in the mirrors as a toddler and you see that the person in the mirrors is you then you you know take your post-it or rouge from your uh forehead and that's supposed to give you a sign that you are basically self-conscious and self-aware but what i want to say is that maybe by at an um and dorsey that adult-centric perspective we tacitly also endorse a sort of like visual spatial perspective on what perceptual awareness is from the um from the very beginning so um while i i was doing my first post talk on in philosophy on uh the sensory motor basis of the self i also started to do a mastering cognitive science and development of psychology and i started to actually address the question of perceptive awareness in a slightly different angle and instead of asking myself what you know what does perceptive awareness is or what self-consciousness is or what self-awareness is i i decided to ask the question how would you really get in contact with the world at most primitive level how do our self-awareness experience or perceptual awareness or experience is more general um emerge from the in in the first place right so are they involved with their lifespan and if you take that perspective right then you realize that actually from the very beginning uh we have that that traditional axis a subject here relates to with an object there is somehow incorrect because in real life the way we start actually perceiving the world is with the subject here that relates with another subject here in a very proximal way right so this means that we'll probably need to shift the perspective in instead of like taking like a tacit

visual spatial understanding of what the perceptual awareness is we need to actually expand it and be more inclusive and uh you know pay attention to the fact that our first experience are basically multi-sensory nature and they also proximal this in a sense like way basically before we meet other people's minds we meet other people's body and why is this important is because there is a huge literature on uh you know what does it mean for uh what what is a minimal self and actually this is important for not only for philosophy but also for um for science and robotics and um many many other fields because it is important to you know correctly define what but really what we really mean by a self and by a minimal self and there's a huge literature um in from from various traditions that you know um talk about minimal self-consciousness minimal self core self uh pre-reflective sense of self um etc and i think if we switch the perspective and take on doors like a more inclusive ecological and bottom-up um approach and what the minimal self is we avoid a fallacy a center conceptual fallacy so for instance i'm trying to depict this fallacy with this picture here so suppose you want to find out what is a minimal model of a self-organizing system such as a plant or a human and then if you take if you endorse an adult centric perspective then you might tempt it to take whatever structure we have as an adult plant or an adult human and squeeze those property or try to you know extra extract the elixir know the structure the basic structure of those property and then you say okay so this is how the minimal self model of that self organizing system is right like you ski squeeze the plant make it smaller and say this is what the minimum model is but in reality we know very well that actually this is not how it looks right so uh for instance if you can see here that a plant at the minimal level you know the the form the stroke structure here is very different that the ones that you have later on in let's say the other life so this means that we need to if we want to extract like you know the basic the skeleton of a certain you know um system then we need a from an ecological and dynamic perspective we need to take into account all these intermediary forms in order to make sure that actually understand what the plant is and what a human is so this is a paper that we as i said we recently uh published this year so we decided to really really go back to um the beginning and to question more carefully this idea of like um um combo treatment and homeostasis uh and i want to start with the disclaimer because this is very much work in progress and there is a lot of work to do because this is as i said is slow it's a shift in perspective so there is a lot of work to do actually to shake a bit you know the the you know the entrance point to the problem and uh this is not an active inference account of co-embodiment and co-homeostasis yet this is work in progress i'm doing this on different levels

from you know um metabolism to you know um more high level level um like self-awareness but because i need to start somewhere we decided with my collaborators basically to use the active inference approach some of the insights of the active inference approach to bring the fundamental state of combat and the co-homeostasis under the spotlight because this is something that we need to do because it's like a snap mainstream so we decided to okay so let's take a sort of like meta level approach use the active inference lens and show that if you if you um like the active inference approach if you endorse it then you we you need to take carefully into consideration this kind of like very low level co-embodiment approach from the from the very beginning so this is what we're going to do in this paper so i'm going to start very briefly with the background and predictive processing framework i'm sure that you are all uh familiar with this uh so i'm not going to spend a lot of time explaining it but i just have two slides on it just because i need to make sure that if someone is looking you watching this and doesn't know what active influence or predictive processing is we are on the same page and then i'm going to spend actually more time and trying to develop the um this and motivate this need to actually take a bottom-up developmental take on what perceptual awareness and experiences in the first place and define the notion of embodiment you know stages of stays in common stasis homeostasis but again i'm going to define them through this um bottom-up developmental lens and i'm going to end with some um hand waving work in progress uh and i'm very open to suggestions uh because as i said this is like very much an open question and we need there is a lot uh more work to do okay so um i'm i i believe if you are all familiar with the predictive processing framework so the basic idea is that our brains is is not a passive sponge that you know automatically somehow you know passively receives information somehow he you know gives the motor output but rather we should conceive the brain as a sort of like active system that anticipates the next sensor input on the basis of prior inputs such as that actually can map the world in a more accurate way this idea very nicely fits with some previous idea that having been developed in parallel by uh the embodied cognition paradigm approaching cognitive science where they basically this author like verily colleagues outline this idea that cognition should be understood as the enactment of the world and the mind on the basis of history a variety of action that the being in the world performs and in the sense like we cannot really understand cognition in isolation from the body and we cannot understand the body in isolation from the environment and we cannot understand either the body or you know the relation between the mind body and the environment without taking into account the history of those like uh connection from the in the first place right because

this is not static phenomena but it's more like dynamic unfolded for good time because this is what life is right so this means that our ongoing perception and cognitive processes and action are influenced by prior events and prior experiences so this is something that you know um embodied cognition approaches and uh predictive processing approaches somehow agree and a different level of explanation but the basic idea is that so you cannot consider this phenomenon in isolation from past events right um so this means so and another i think common ground is this idea that humans are biological agents that emerge as like proactive survival enabled prediction machines which are strive to reduce uncertainty within a highly volatile environment so we basically try to make sense of the environment out there because you know the world is bigger than our heads and we try to find a way to actually navigate this bigger world which is highly unpredictable and potentially threatening so one idea is that in order to keep track of this survival and reproduction relevant bodily and worldly information our brain solves that problem of uncertainty by generating self and world models and it does so by extracting statistical patterns of information from its embodied and worldly interaction and there is a mathematical story behind it i'm not a mathematician i'm a philosopher some you know um leave that to the mathematician but i think what i need as a philosopher i need just this idea that you know our ongoing perception cognition and action and process through the lines of prior self and well-related information right so this is what i need the only information i need the only thesis i need now if this is true then this means that uh it becomes crucial to look at how this experiences an action perception gets off the ground from the outset right because we need that lens to be accurate right in a dynamic and ecological way one basic and overlooked aspect of our embodied and predictive processing approaches in both philosophy and cognitive neuroscience is that brains and minds and human bodies first develop within another human body uh and this is important point because crucially while not all humans will have the experience of being pregnant or carrying a baby the experience of being caring and growing within another person is universal so this is something that all humans are concerned with right so and this is something important as well because way before we meet other people's minds we literally meet on to logically meet other people's bodies and there's something i developed um already into 2017 where i looked at empathy and basic form of shared experience but the idea is the same it's like okay so if we really want to understand the relation between uh mind body um and how um you know embodied creature relate with other with others and the world we really need to go back to this basic idea that yeah before we meet other people's minds we literally meet other people's body

so there is a lot of work previous work that show that looked at the homeostatic and bodily mechanism operating at early stage of developing in infancy uh so uh post a birth right uh we decided to take uh a step further and to focus on pregnancy uh because these are uh this is a universal case remember i i i i'd like to emphasize this because typically you know when people hear the word pregnancy say okay this is a woman's stuff uh because well i i mean certain certain people certain person can get pregnant other don't so but if you take the perspective the other perspective then you realize that actually all humans are concerned with this experience right where two individual organisms literally grasp and grip into each other and um contrary to the common view that the fetus is passively contained and sort sometimes just put us like solipsistically trapped within the solitude and the silence of the womb uh we will present evidence showing that actually uh there's a lot of going on there there is an active in bi-directional co-regulation between the two living bodies and to live in organism um and we need to start with some basic definitions so first of all we're going to talk a lot about embodiment but we need to make clear what exactly we mean by embodiment in the first place uh there is a lot of work already in the literature under the relation between the body itself um i highly recommend this like um very recent uh review by Keenan colleagues coming from York Notes for Love they have a very rich um overview of you know the relation between the body itself and self-awareness uh but here i'm going to use um my analytical philosophy card and i'm going to define embodiment in its minimal form to refer to the instantiation of body properties by a given individual in our case a human in other words old talk about embodiment can be conceptually ontologically linked to talk about embodied beings and then and their bodily properties now the key starting idea here is that brain and minds as i said it's like if you're taking the body cognition approach and also more recently like the predictive process or active inference approach it's like they all seem to agree that the brain and minds don't emerge in a vacuum but develop and evolve as a system dedicated to controlling and observe the organic survival and reproduction so basically if you take the classical mind-body problem instead of conceiving the body as a sort of like a vehicle or car that you know is it's there to you know support and fuel the mind actually you should consider that way around namely that the mind has evolved to actually subserve the needs of a uh organism right uh of a body um of your human body so you basically shift the perspective which one which is actually which comes first which one is actually more important so crucially however if so if we endorse this perspective and we agree with this shift then we need to consider the fact that human bodies do not emerge in a vacuum either right so human bodies basically emerge and

develop within another human body so this means that we need to adapt and extend the notion of audimen to reflect this fundamental body within a body case which again i'm going to stress again and again this is universal right it's not constant just like a part of the population so but what does it mean to say that the two agents are co-voted well uh in biological terms this means that uh as we'll see later very shortly is that the regulatory of the two agent states and particularly the need to maintain that by physiological stability despite the fluctuating and unpredictable environment is actively negotiated between the two organisms that share for a given amount of time typically nine months but could be you know shorter a bit longer common and bodily and environmental uh resources right and i'm i need to give you an example so what exactly i mean by going um because i think this is a key point so i'm going to ask you to walk with me into a thought experiment because i'm philosopher so i'm going to present you with the thought experiment so you imagine you um you have a friend and you uh you know you you have to perform a sequel show right and with your friend and you have to share this pair of trousers here so your friend puts his his left leg in the left part of the show of the trousers and you put the right leg on the right part of the throttle uh throughout this and you need to coordinate your walking such as actually you can walk as fast as you can to actually reach a target right an apple stay um now uh suppose that your friends as hitting his toe against rock stumbles and falls and experienced them by fear so what is happening in his body in his organism so um this means that maybe your friend is going to untrain you in full in which case you can say that the two bodies i mean you the friends are called fully or you shared a state of like falling together and you can also say that well because it's kind of like negative events your friends your friend is kind of like experiencing fear and you also experiencing fear as well while falling so you share that state of um you know fear not crucially however that this is not necessarily so so suppose for instance that you know your friend is uh is a master yoga teacher right uh and while you are following you your friend is somehow you know managed to keep the balance and or the you know the tissue of the trousers kind of like you know destroyed and you know one is standing at other one is falling right it's not necessary so you can imagine in case that actually this is happening and also imagine a case in which let's say well why your friend is falling and experiencing fear you find the entire situation actually pretty funny and instead of experiencing fear you experience amusement right so the two emotional states are not necessarily shared so your friend is experiencing fear or your um experience like say amusement so that's one example of like what we can understand by sharing a state now let us go back to the case of pregnancy right

not that the case of pregnancy is very different because if you take exactly the same example uh and say that the mother is like some stumps under on a penis something and you know false then there is no way in which defeaters can say wait a minute actually i'm not i don't like the situation i don't want to fall i don't want to experience fear and i don't want to experience negative you know emotion i'm going to sit you know stay put here you go ahead for right so this means that actually there is a way in which the two states are the two organisms necessarily such as the body itself regulatory processes of one organism necessarily is linked to the um the other one so it is important to distinguish the case of pregnancy from the case of two bodies like contingently happen and to share a state right and we can just define the notion of like co-embodiment and saying that two or more if you have twins or uh more more than three to two uh kids are co-avoided in this strong case if they are bound to share bodily and environmental resources for a set time period typically nine months for survival purposes uh note that pregnancy is not only a case where we can find instances of this type of like stroke or strong embodiment like foreign siblings but we decided to focus on the case of pregnancy because as a status is universal case all humans have you know experienced this so there is already work on the fascinating phenomenology of shared embodiment and i highly recommend to to read it in what follows basically i'm going to very quickly uh go to start with a very basic level of actually what is happening at the you know from single cell going up to the organism and that kind of like shared um shared embodiment again this is fascinating a huge literature when you think about it we start with like this tiny little round thing with another person's body and then we you know become ourselves um so uh again i'm going to be very quick and very sketchy on this like fascinating pro you know process but i'll i'll try to flag you know the the crucial steps and the crucial stages right uh so the first stage is the process of implantation so basically when there is a dialogue between the fertilized egg and the mother body right and this is actually this is the crucial step of a pregnancy a successful pregnancy um because for instance most of the miscarriages happen during this step um this is a like a sort of like negotiation process going on between the two organisms saying uh and like any negotiation things can go okay or can go wrong right so this means that both the blast to see the baby and the endometrial tissues they need to actively regulate gene expression transcription factors signal pathways inhibiting factors and growth factors during that implantation process this is highly complex process and very fragile and very delicate if something goes wrong it can go to miscarriage or um later on and it pulls at this stage the maternal body literally self modifies some of through differentiation into specific type that

support implantation right so the key idea here is that embryo and the mother's body need to coordinate with this cells within that very tiny implantation window right now if this process let's suppose is like successful then the coordination goes uh to the next step between week seven and thirteen more movements um it involves self-organization can start so and um and the self-organization um uh interestingly involves movement which is actually very important because especially for the active inference community because as you say as you can see actually movement is present very early on in our lives so these movements are initiated from the so-called uh central pattern generator in the early spinal cord which produces around with eight very abrupt startles or the entire fetal body right importantly note that this throttles which is somehow automatic right as we think of like a sort of like heartbeat right it's going you know automatically uh provide the minimal basis for the most basic occasion of sensory feedback because in doing so for instance and might start to touch the face and that's softly how you have my hand you know autumn automatically or unwilling to attach my face and then i received this like sensory feedback on my face from my right hand or my leg might change its position be flexed or extended or both so this means that i might have some minimal sensory feedback at the stage that that fools the possibility to respond and adapt to these movements this means that self-organization and fetal movement initiate as a biomechanical process that progressively builds sensory and regulatory capacities so for example as i said by being stimulated by startup the fetus perceives that the hand is movable and that touching the face rather than the uterine wall with hands uh might feel different because i'm more sensitive here on the face that actually i can i can feel my i can feel my fingertips so if i touch the uterine wall but if i touch my own face i have a double touch right okay so with this at hand we can go further to this notion of homeostasis and bear in mind that this is all happening so we have a developing body within another body now homeostasis typically refers to the organism's need to maintain unstable organic internal environment despite to fluctuate in external environment right and the homeostatic regulation typically entails using feedback control mechanisms to keep the senses viable of that internal environment within the ranges compatible for the survival in this case for compatible human survival now it is well established that this control involves uh automatic actions like motor reflexes endocrine immuno immunological and autonomic processes um that are driven by the feedback inputs and the resulting prediction error which is the discrepancy between the expected bodily states uh at a homeostatic set point and its actual level as signaled by the sensory inputs from the body uh and as early as 1929 uh it has been you know proposed that

uh maybe the control body homeostasis consists to the primary aim of a brain function right and this is again this uh underlying exception is that the fundamental drive towards like self-preservation to take care of this body this self is essentially linked with constraints of acting within the body's visual states right so it has been classically proposed to define the homeostatic set points at the expectation the means of prior belief about the states the body should inhabit uh and or homeostatic range as the variants of this are prior beliefs now going back to our case right so let's go back to the to you know just for one or the primitive level so many years it has been assumed that the fetus has no metabolism of his own right uh now is nowadays is well established that actually the placental and the fetal liver working on them as a sort of coordinating multi-organ system to provide unnecessary nutrients serving the fetal metabolism and growth and this is important because you know if you talk about homeostasis we need to talk about metabolite because glucose is the most important fuel for fetal metabolism and ultimately for homeostasis so this means that maintaining an optimal balance as a continuous supplies of nutrients from the mother to the fetus is critical from the development and from the fetus standpoint is basically critical for maintaining his own um you know homeostatic balance now note that the fetus may be able to control and save some energy to some extent through many complex and dynamic processes but importantly however it is oversupply and under supplied of nutrients by the mother right uh which are process controlled by the pregnant person that can permanently uh program the fetal metabolism adversely right so this means um um the the way the mother um and handles her own um metabolic processes significantly impacts the way you know uh the babies um you know has his uh self-regulatory process maintained and there is accumulating evidence that there is a direct link between mother's adverse bodily and mental health states like depression and anxiety during and child medium mental and physical development but this is this is not only for this is valid not only for the negative states but also more like neutral states such as food preferences and flavors that are passed on uh from the mother to the infant so there is a strong strong codependence there now the the key into the into the intermediary conclusion at the stage is that the emerging self-organizing organism uh human organism is not only self-centered but also crucially co-embodied and co-depending upon homeostatic processes within the environment right so that's the basic um thesis up now now we need to take this a step further we need to talk about also action anticipation and not just like our prediction or perception so we saw up to now that homeostatic regulation requires a model that enables inference perception and prediction action selection importantly however the brain initiates

regulatory responses prior to romeo city alteration provided it can be anticipated so this means that you basically act to go to the fridge to seek food uh because you are hungry but actually you aren't you're not you you're not uh wait you're not waiting for the starvation to happen to actually go to the fridge you do it well before that actually is going to happen so you anticipate to that state so this means that higher level that enable prospective control are leveraged with two essential components like inference and prediction which brings us to allostasis which is defined as anticipatory omniscient control basically you take action into the world because you need to regulate your inner homeostatic you know balance but you're doing this by taking action into the world and importantly you're doing this by um anticipating you know that state you don't really expect to be on the edge of like you know dying from starvation to actually take that action if you are optimal you know um surviving um organism so this means that the process whereby the agent selects action that will most probably bring about the desired sensory entries while explicitly or implicitly modifying the true causal structure of the switch to the guaranteed recurrence of those inputs over multiple time steps um so this means that i can act differently in different environments i can go uh to a fridge if i'm hungry uh but if i'm in the desert i could you know if i expect to see fridge uh and you know i don't see it uh then i'm going to probably die instead of looking for a fridge i should probably look for i don't know banana tree or coconut tree right so the environment defines what type of expectations you should have in a particular case now there are two points that are important here note that uh for the individual living organisms homeostatic i mean we can conceptually distinguish homeostatic processes from allostatic processes but in real life those go hand in hand right to the other two faces of the same coin this is because we uh humans we are not static isolated creatures we are dynamic beings depending on the rich environmental resources and we depend upon social interaction to achieve survival and reproduction and to give it like the basic example that i really like is that like the breathing like um it's it's a perception so it's an open-ended process right we are constantly exchanging information with environment via breathing right uh and this is a dynamic process because my body moves right my thoracic cage is moving all the time and this means that those type of processes like um allostatic processes and homeostatic processes even though we can conceptually distinguish it in reality they are actually coupled this means that perception in action homeostatic and allostatic self-regulation are coupled strongly coupled in a dynamic load that keeps the organism permeable to and connected with this surroundings for better and for worse because when i breathe in i can take in the air right uh that is necessary oxygen for my survival but i can also

get in i take in the coronavirus you know and you eventually potentially die right so we are open system for the better and for us in that sense right um the second point is that uh going back to the our case here is in pregnancy is that self-regulatory architecture is multi-layered this is an interesting point here right it's like it's already complicated for individual but imagine for actually two individuals that are literally gripped into each other right because the baby's regulatory processes are coupled with the mother's homeostatic and anaesthetic processes and this is achieved mainly through the placenta which is an fmo and intermediary or that enables vital and biological exchanges between the two bodies for example it allows the infant to breathe despite the lack of proper proper lung regulation and ensure that the infants is fed despite lacking proper eating effectors not again that placenta is the relational organ excellence but is also a universal organ whereas typically people associate the center with like the female body and a certain type of population to say to a certain type of persons but we all have use all humans have used successfully this organ in order to survive and you know i don't know do philosophy cognitive science and active influence later on in their lives right and this is important to keep in mind now if i'm right this means that a dynamic and complex system such as the human body needs to be able to play a double game in order to survive and potentially reproduce it has to successfully maintain sensory states within a certain physiological balance but it also has to flexibly change the state in order to adapt a constantly changing environment so if we look at the human body through through this dynamic lens it becomes obvious that what happens in between the organisms environment i mean the boundaries between the two play a key role in making sure this game is correctly played and flexible enough to actually maintain the organ is alive uh so this means that we need to define the krishna uh critical notion of boundary or in between right we need a new conceptual toolbox in order to make sure that we get that process correctly right and this is something i'm currently working on with some collaborators and some some of you probably know that you know the notion of marco blanton uh marco blanc has been uh you know advocate as potential candidates to you know promising way of conceptualizing this key notion of the boundary the medium between the two and again this is like uh uh working progress and there is like um a lot of work to be done on that side but for for this talk i i really need to retain this like very simple idea that homeostatic and anticipatory self-regulation depends not only on the body states uh and but also on the external and more crucially on the relation between the two so we can just define co-homeostasis as the bi-directional process of co-regulation that happens in between two combated organisms and homeostasis involves self-regulation through others couplet and

homeostatic and aerostatic loopy states not that and this is something that we have to you know um you know work a bit with the reviewers on our paper and they were kind of like right pressing clarification on this point because it's important to bear in mind if we talk about bi-directional process doesn't mean there is a asymmetry there or absence of conflict on the country actually things can go wrong right um in the sense like uh it can be that you know one uh one agent actually takes the balance too much in one sense than other and to give you an example like what exactly what i mean by that so think of a uh highway right uh so you can have that's that's a bi-directional highway traveling from north to south let's say that you have typically more cars coming from north to the south from the mother to the infant but you have if you can have one single car traveling from south north uh and that car is like you know i don't know carrying bomb then actually that's irrelevant right it's not the quantity that matters that actually the quality of the information that is carried between the two organisms actually have many many cases and basically during the pregnancy you know mothers our pregnant person develop um diabetes or other um immuno immune immunological disorders right uh so that's not necessarily uh a straightforward process as i said and i'm emphasizing this this is a constant negotiation between the two organisms it's a dynamic open-ended process that can go okay but can also get go wrong now the big claim the radical claim we want to make is that given the necessary and universal coinboardman sustaining the human body's self-regulation and self-organizatory processes from the outset the early homeostatic regulation should be fundamentally considered as co-regulation hence communist thesis now uh in a further work we want to develop this idea of uh when exactly and how the brain starts to build a self and world model in the womb right uh because it's like it's a key question from uh for the you know predictive processing framework and active inference community so how exactly this is happening when exactly this is happening this is highly speculative um and very much work in progress um so it's it's an open question whether the fetus is able to generate self and oil models and if yes at what stage of development some people argue that this might be possible only after the basic talmud cortical connection are established in the fetal brain around 20 weeks of gestation this is because so the the main reason is because after this time point sensory information can be transferred to the fetal brain uh before they establish of this uh telomere cortical connection the fetus just acts on reflex automatic mode however other people uh argue that through the central part generating seemingly chaotic exploitative movements they also um can generate a sort of like a sensory motor coupling and or some very basic primitive sensory experiences um again this is like an open uh question and this is something

that i address um in a preprint currently under review we are two of my collaborator uh adam safron and jonathan delafield uh we look at how exactly self-awareness no so how exactly the fact that our first experiences are co-embodied impact may impact the way we understand current theories of conscious experiences in general and what we our provocative take is that you know our consciousness study focused too much on visual experiences uh and a visual spatial way uh perspective endorsing a tacit adult centric and visual spatial and we want to say that perhaps conscious experience are not all about seeing red apples to do you know famous example so what are the implications of switching the perspective you know from this like what we understand by minimum self and minimal self models you know from endorsing more ecological bottom up rather than adult centric and um vision-based perspective again this is very much work in progress something that we are working uh on right now but you know just like bullet points uh if we switch to perspective then we need to question the individually static and solipsistic approach of understanding of the perceptual link between the ancient environment we also need to question the adult-centric thesis of sort of like fully-fledged emerging individuals like you know athena greek god or this athena coming out from zeus head so rather than humans emerge within other humans and develop through close interaction with other humans uh we also need to question the idea of a self-awareness for minimal self developing independently of other cells this is because as i hopefully you know uh convinced you that we basically were ontologically depending from the others at the very basic biological level and then the question we need to also question the idea of a perception being a solitary and distant deva like seeing a red apple uh but rather we need to you know see a perception as an active interaction active engagement with environment like for instance touching human being or coupling the interceptive like signaling like the breathing so the way i feel the air inside my body the where i take the air from outside um and also important that perception of that perception you know that perception comes before the seeing the red um apple and if this approach is right this means that that early primitive perception influences the way we perceive and see the red apple later on in life and to take home messages that uh when it comes to understanding nature of our perceptual experience in general i also have an argument for this in this paper but um you know we are trying to build a case for this idea as well and conscious experience as well um the infant is father to the men uh to the human sorry uh to paraphrase this um metaphor by the english poet waltwood and this is it i would like to thank my collaborators for working with me on radical project uh i would like to thank uh organizing of the active inference lab for inviting me uh i also i would like to ask you that if

you are interested to collaborate with me along those lines of like paradigm shift please drop me an email because i'm um very open to these ideas and to develop further um and yeah this is it and thank you for your attention thanks ana that was really interesting maybe you can unshare and we'll have a little bit of time to discuss cool so if anybody wants to post a question they're free to do so i have a bunch of thoughts in question but one piece that right at the beginning you mentioned that the historical nature of inaction and it just shows that sometimes when people are talking about inactive approaches they're oh you know picking up this pen it's the snapshot inactivist but actually the implications of action are that we have to think about history and development so i really appreciate the way that you framed it in in these terms um so maybe just to get things started where does active inference come into play here that's pretty big question but where does active reference come into play yeah so i think active influence come into play in the way we actually understand how the minimal self is constrained right so how the minimal self organizing system is somehow negotiate that exchange of information between you know his own self model and the environment so the radical idea we want to put forward there is that the closest environment and the most primitive environment that we have in is another human body is not some sort of like objective environment but another subjective i mean lived up lived you know uh body right uh so i think that's kind of like radical idea and this means that we basically we need to not add an extra level even more dynamic to the active inference to say it's just not enough to basically just act and to make sense of that action in the world and the movement but actually we need to take it as a sort of like double way it's a core coordination uh between the two organisms right so it's not just like function it's just like co-action and core movement uh so this means that it's even more messier than we thought at the very beginning it's like if you take you know the the standard uh approach right so you have the environment the agent and then the agent needs to build the model of that environment basically by contrasting you know whatever the self internal self mother says and whatever extracting the mapping of the environment has right now you need to take into account that that's a kind of like environment it's not only just like and a another agent out there that you need to interact with and that agent can be i don't know another human being or a robot right but actually you share common resources computational resources with that another agent right and you need to actively negotiate those resources on the fly right even though there is kind of like a um asymmetry because obviously the pregnant person has way more power uh you know she can uh he can move around and um you know act in the world but uh that co-coordination happens from that makes me wonder how is it

related to niche construction that's something Axel Constant has of course worked a lot on so how do we think about niche construction is there a niche a shared niche is one the niche of the other yeah so that's that's i mean this is like as i said this is an emerging field and i have something like a zillion questions an interesting question to us and one of those is like related to the niche constructions for instance of this like homebody so the the pregnant person right so there is like very nice work showing that uh because also the mother is embedded in a wider you know social and physical environment so we have cultural right constraints on that pregnant person and there is work showing that you know for instance like um you know the the the maternal language for instance influences you know the the way the infant will later on cry and if and also if the niche so to speak for instance is bilingual because i don't know the father speaks one language and the mother speaks another language and also for instance the the infant later on cries or you know coordinates right so this but again this is emerging evidence we need way more work to actually carefully look at that exchanges between you know uh the embodied subject the core body subject the pregnant subject and it's like close environment but what we know for sure is that clearly i mean um i mean the the the the social background the cultural background and the physical background is really really important right has direct direct impact right and actually there are there are studies that show i can't remember the on the top of my head the name of the authors but i yeah they show quite nicely that actually you know uh uh i don't know french babies or german babies they don't cry the same way because the language spoken by them the mothers has different intonation right uh so the auditory input that you can have uh in the room actually significantly impacts right so this means that if you change niche the cultural nature change environment you're also going to change the way the baby later on will somehow express you know this regulatory process like crying right i'm not happy right that makes me think about how even as adults we're still bounded by externally constructed membranes we rely on a broader society to bring us glucose and to take our waste and so there's a critical period just like you had with the seed absolutely a critical period where just like for any other complex system it's going to have bifurcations and a lot of leverage points early in its development when it's literally more fluid but later on these things are different in kind of uh detail but not in type so extremely yeah and i think it's it's really important because we we tend to we need to i mean i i forgot for that slide but i think it's important to bear in mind what we call the invisible basis of the aids ice bank right so certain things we really take for granted so if you go go again to the plant metaphor so um or the tree metaphor so think of a tree right so you walk on a street so let's suppose you're a

martian right and you land on the earth and you see a tree and understand how the tree works what is a tree right so if you limit yourself to only to what you see above the surface of the earth right like the trunk the leaves you know um then you might miss some important bits of information of how exactly tree works because the tree also has roots and basically the way is negotiate the nutrients through that basic roots invisible ones you know significantly impacts how later on i don't know the trunk or the leaves are constructed right so we really need to this is our motivation to to endorse this like bottom-up developmental and ecological dynamical perspective because we really need to go back to the roots invisible one that we take for granted because as we said rightly point out maybe there is something going on at that critical stage that if you know act upon that critical stage then you avoid later on bigger problems right that you can't modify um so you need to you know act exactly at the time t for a very specific reason but also those roots are highly connected with i don't know the nutrients in the earth in this case like i know mother's body which the mother body is kind of like connected again with the closest environment i mean the way he is like supported or the cultural background that the type of support she gets from her close environment later on and then you know i don't know going higher up and it's like policies that you know governments can have and you know climate change and other things right so it's everything is somehow a very open and uh precarious system and i think this is i mean the way i understand and this is why i'm interested in this like entire you know new active inference approaches because well as the name shows is like it's an active thing right it's like it's constantly dynamically moving then we need to make sense of it right in a very um you know rigorous way but still you know um it's it's an open-ended process right yep and for the tree visually the roots are there but with humans our roots are often transient or they're cultural so it also asks us to move beyond a visual representation of being tangled in a web and you can't move but rather to see these dynamical processes that um go through the developing embryo so here's a question from the chat um anna asks saying that they just came to see your talk after hearing about it just a few minutes before it began so that's so fun that we can connect with people who are just hearing about these ideas maybe for the first time and the question is um i would like to ask for background references to start studying this theme so what would be some key references or some key questions or topics to look up for somebody who's kind of wanting to follow you on that freeway on that research trajectory that you're on so um so there are several so it depends on the background right so uh uh there is there is very nice work on the phenology side so in philosophy right so there is a nice work by um alaina by and also by honeydew and um i think

i i i hope i'm not misspelling her name but if you google hannibal she has a recently last year paper with i'm sorry i'm really bad with names um so that's a very good reference starting point uh i also wrote on this uh so if you google my name i started to work on this already in i i worked on a basic form of self-awareness right um and also there is nice work on um um the the touch and effective touch component right by uh maria laura philippe and laura cuccinelli they have a very nice review paper and a special issue i edited it is called such a special issue called the relationship basic forms of self awareness and they have a paper there and actually there are quite nice papers there as well on this topic uh but more overall i wanted to say that you know i have um the twitter handle uh public and there is um i also have my website and uh the email address so i'm very happy to take uh emails and i can basically afford um a normal um you know complete reference because on the top of my head i'm pretty sure i left aside important references and i don't want to you know say i i you know i said this references i forgot those references i want to you know keep diplomatically so email me and i'm going to send you references what's interesting uh and it reminds me of the way that now the internet is allowing us to be in very tight feedback loops more like a co-embodiment of knowledge you're able to share your knowledge and your perspective on things and then somebody can ask a question and now we're in a very tight proximal communications wise cycle where we can actually uh update each other on oh it was actually this 2017 citation she was talking about not the other one but if that was only uh once in a while conversation we couldn't actually develop at this rate so that's quite interesting again as adults who are co-constructing each other's ecological niche um okay one thought i had and if anyone can write more questions is in most insects so as an entomologist was thinking of insects most insects lay eggs and you had said that um the placenta and the co-embodiment of pregnancy is universal and maybe it's mammalian in fact it's defined in humans oh yes yeah yeah universal to humans yeah so the egg for the insect contains everything it needs to get to the relevant life stage like to hatch from the egg and then be whatever stage of development it's in um except for the ants well there's other um exceptions here but in the ants the larva is um after the egg it hatches into a larva very quickly and then the larva is actually in a two-directional relationship and the larva are called the social gut and they have a one-ended gut and so they're being fed food and actually digesting it and in a two-directional hormonal relationship and so it was so interesting that it's almost like the length of the dialogue that the the length of the intergenerational transfer is it one day is it one year is it 18 years is it permanent are we always co-creating the length of that handoff and the richness of the handoff seems to have something

to do with the evolution of complexity so i was wondering what you thought about that yeah so this is amazing i mean i i mean when you look at nature it's like mind-blowing i mean i i i was about to say it's like we i mean i'm talking from a philosophy perspective i can safely say that your philosophy has been kind of like adult-centric and you know uh you know human-centered anthropocentric right um uh but if you really understand this like how you know mind and body works and you look at the the nature more general then you realize wow this is really fascinating and so um you know we can learn so much so you you are telling me about the ants but i i recently read something because i started to be fascinated by this idea of like placenta right and i've read that for instance some lizards right so if you have exactly the same species of lizard right and some of them lay eggs and some of them actually have babies exactly the same lizard and you ask yourself why and then you realize that actually it depends on the climate where they are if they are warm climate or a cold climate right so if they are in warm climate they lay eggs because the sun is kind of like you know but if you are in a cold climate then you're going to keep the egg inside and make the baby protect the baby inside right and then only after that baby kind of they safely put it into the world right so that's a very nice example of as you said it's like the the flexibility and adaptability that we can have to actually know the make life happen and carry on right because in this case you basically have exactly the same body um but the way you carry on life is depending on exactly that in between interaction that you have with environment right in a cold environment in one case and warm environment another another and that impacts the way you basically call you know the way you you know um self-organize and reproduce it is really fascinating i really hope that basically you know um animal scientists and you human brain scientists would work hand in hand because i'm pretty sure will somehow unlock many of the so-called mystery of the human mind and human body just by looking carefully what is going on to you know uh friends and neighbors right animals right it's an excellent point and in animal cognition especially it's phrased as like the development of the cognitive capacity as if that was the end point you know you go from zero to 60 and then you're at 60 and then you cognitive decline you know tap on the brakes but there's a richness and it's different cognitive agents which actually brought me to this question which was um i was definitely with you that the adult centric perspective that you raised in your first anecdote at the um conference and it's oh it's all adults in the room so let's go with this that it would be fallacious to have a adults-only perspective and maybe it might be also somewhat of an error to have like a child only philosophy just whatever the child appeals to them let's just go with it so how do we structure

in the intergenerational conversation and learn from each other yeah so that's that's a that's a big so i i think for what we really wanted to do in this paper was like something very very modest namely to draw attention on the fact hey guys pay attention we have this lens yeah it's kind of like tacit let's clean it a bit and expand it right so i'm happy to basically expanded it and take that kind of like bottom-up perspective but obviously i don't want to bring everything to the to the root so to speak right would be like exactly the same fallacy to like focus just like like on the fruits like the apple in the tree and to focus on just on the roots right i think what really needs is an explanation of that dynamic process how that unfolds how that emerged how is connected with the wider environment and with other minds other bodies other animals um i don't know if it's true i need to check but at certain point somebody told me i think i saw it on twitter that actually if you take a human body and you take the cells and the human body numerically actually have more foreign cells than actually human cells in your body numerically because all you have was my microbiota anything right so if you if you take just like the quantitatively then we basically are more than another than actually myself which is like i i need to check that information if that's true that's actually quite interesting one one thought on that um number of cells is actually the co-embodiment framework moves us beyond a numbers game for example there could be a very big system and a small system like the earth and the moon and yet the moon is relevant for this or that process on earth and that's because they don't have to be symmetrical if it was a tug of war then it enforces a symmetry which enforces a narrowness but actually it it doesn't really what if it was really influential cells or less influential cells that can be accommodated in an active inference um or in a dialogue based framework but less so in a sort of there's only one way to collaborate like tit for tat or game theory which is where a lot of the models of pregnancy uh just are still scratching their head right um here's a little of a meta science if anyone else has any they can post to is it was really cool how you um demarcated your thoughts on being a philosopher and working with a formal theory so what are your thoughts on working with formalisms as a philosopher or working on these transdisciplinary teams where there might be colleagues with clinical psychological math and a philosophy background so how do we work together on these teams and address these topics that are really big yeah that's that's a that's a good question and sometimes it's like twitter is quite good for that because it's like he just wants me to get this like the this conflict in the nutshell so i remember somebody tweeted at some point you know when there's all this computational approach is kind of like emerging uh and uh regarding like the clinical population they were saying bad news for clinicians they would

have learn mathematics and statistics right to build computational models right and which somebody replied i said bad news for mathematicians and statisticians they will have to go to hospitals and talk to the patients right uh so there is i i like that kind of exchange because this is the problem that we have in the sense like we really need a dialogue between you know the world out there you need to go there talk to the patient gather the data observe the insects you know observe the human right you know um spend countless you know um hours is like measuring how many eggs you know what they're laying and everything uh and in the same time you also need something more rigorous that will allow you to basically you know build models that you know you know instead of like how should i put this is like as if you you you you test hypothesis instead of a testimony in real life you you let mathematics actually take the risk of that hypothesis and if it's strong that's fine because it's not going to die it's just a wrong hypothesis but in real life if you test hypothesis and you're wrong then actually you you may die right so i think that's should be a very nice dialogue between the two uh two approaches um should be seen like a sort of like dig in the same tunnel but from the toe opposites sides and hopefully meeting in in the middle uh personally i like so i'm i like to even though i have a philosophy background i like to consider myself as an interdisciplinary researchers so i have collaborators from all fields i collaborate with clinicians i call i'm doing experimental science so i'm running studies i'm collecting data i also collaborate with people on the more formal side so we're building kind of like you know um simulation and formal approaches to a certain certain phenomena especially connected with my main interests which is self-awareness minimal self-worth alteration of minimal self-worth um the relationship between the self and body and co-embodiment and um so on and so forth but uh but yeah i i do and this is why this is where i said to the last in the last part of my talk in the last slide that i'm very open to collaboration uh especially on that side because i really need to i don't really think we reached a point where we basically need to have a dialogue because it's like uh we have so much information and we realized that this life out there the mind and the body and everything is so rich that we cannot we cannot solve it just like we just like one single discipline or one single tool we really need to you know join forces and um a dialogue it's going to be a bit more messier a bit more trickier because we don't have the same conceptual toolbox so we'll have to do a lot of work beforehand so to clarify what exactly we mean by that right in different fields and i think this is where philosophy can be useful because we kind of like obsessed with this idea what do you mean by you know give me a definition to make sure that actually we're we're not all over the map you know we're talking the same phenomena about the same

phenomena but also i think it would be dangerous for philosophy to somehow develop this conceptual toolbox completely cut off from whatever you know you know the the things out there the empirical world or you know the mathematical models actually showed us right so we need to keep those gates open and um you know exchange and dialogue and collaborate wow thanks for that awesome response um to call back to your beginning slides you had the armchair philosopher and that's a term that people know just like uh in america we say monday morning quarterback just like looking back at the past and um the armchair philosopher is not so different right and that made me think of i think therefore i am and the idea that by actually getting into some sort of altered state of inference by meditating or by doing some other um thought ritual it would be possible to reach insight through thought and the way that you just framed the answer was like hypothesis testing is an action that we take in the world it's not just a thought and when we frame hypothesis testing as action about the systems that we care about like the next generations then it makes us think about working on teams and on accountability and on reproducible science all these things that are sort of like on the side of the armchair philosopher and the armchair philosopher might also care about participation in science but that would be a secondary conclusion whereas if hypothesis testing involves stakeholders and it's also uh something that's happening in the loop it's a it's a different situation so it's just really like interesting i hope for those who are listening to hear what you're saying uh and actually i keep i i like to keep reminding my students that you know i i put a picture of the card because i'm french and he's the his friend just like the counter anyway but i could have been aristotle but i i don't know it's like those kind of like classic philosophy we need to keep in mind that these people are i mean aristotle was like biologists right he was like constantly dissecting things and you know testing things just was like looking at what was going on with the organism so aristotle and he's considered to be a great philosopher and similarly with uh with the god i mean he was like a mathematician right and he wrote a treaty on optics right so it's like a vision and everything else so so um science and philosophy were kind of like working hand in hand and i still think they're actually working they factor like i really think that actually scientists they doing without realizing actually uh and uh some good philosophers actually they're keeping an eye or whatever is like scientists are discovering whatever nature is telling us right um because it's like this is a constant information processing writer we never get to the bottom rock of the reality right things are changing right it's like and i think what is really fascinating with this um example of the virus right let us let's take the virus because this is kind of like very recent and very rarely invented

and very timely right now so if you are a classical armchair philosopher then you have a huge ontological puzzle with virus because it's like this is where it's keeping keep constantly mutating right and function of whoever is hosting him right uh or you know whoever the host of that virus is creates different type of mutation right uh and this is like it's an ontological puzzle for uh philosophies like what is the virus which is like coronavirus which one is like it's like and you you feel like saying was actually none of them and all of them right because somehow all that kind of like share this type of information but it's changing in function of whichever body of number of bodies and that depends on i don't know the culture and the food they're ingesting or i don't know different genetical components and so many other factors and i i i can literally say that we can literally see that actually this is like we literally new beings coming you know new new entities arising like mutants mutant forms of like life it's like constantly arising so quickly right that we can't even you know wait a minute let figure out what is going on right uh so i think we have a very nice and humble example example of how reality out there constantly is coming to us with like constant flow of like emerging information emerging creatures living being self-organizing systems and sometimes it's like well we can't seem that but actually they kind of like dangerous they can kill us uh with all our science you know um and that's i think that's a very important point we need to bear in mind yeah thanks for that it it made me think that it's not just emerging entities it's actually emerging interfaces and even if the number of entities were set the interfaces would still be combinatorial it would be explosions just like words even if the word vocabulary is set you still have interfaces that can emerge and then um as we say also here in the putting descartes before the horse absolutely an error um but then it's actually kind of something that tells two layers of truth because it's philosophy and science that are working together and how we think about things and how we operate things and what do we hear right now other than that there's a disconnection between technology and society information flow governance decision making all these things you just you know spin around and pick a number and it will be related to something like that and so that suggests that there is a requirement for a new kind of a conversation yes i fully agree do you have any other thoughts otherwise this was really an awesome guest stream and we all look forward to hearing from you in the future but yeah you're always welcome you can come on whether you're the author or not yeah let's hope that you know uh this will kind of like develop more uh systematically and uh when i when the situation allows to basically you know uh organize a platform or perhaps uh a workshop in person meeting such as we can actually you know put this idea to work because i think that um

uh just kind of like uh proximal face-to-face interactions have a different can can have a type of outcome that you know the ones we can have right now uh so we can you know consolidate the collaboration dialogue on um on a more uh uh you know concrete and pragmatic basis uh what i really i realized that during the pandemic which was really really fabulous is that well uh as what's happening right now it never occurred to me that basically well well wait a minute we can have like a zoom conversation with somebody from california talking about this so just kind of like um you know crisis um you know brought up the surface new way of connecting right and make us um richer in a way but i also think that it's important to have that you know range of problems in place in the first place to keep the balance right right uh because we have evolved going back to the human nature and human organisms we have evolved to basically move and interact if we don't move we don't interact we simply die and uh we need to you know keep moving and interacting with each other and uh i think that's that's also um valid for different disciplines that's what we basically need to you know adapt and interact with each other as disciplines as active lab in online participatory lab we will co-embody that event in person when your region is available and when the personnel are there then we'll scaffold it in a new way because previously it was all in person there was never seen a need to live stream or never seen a need to document to translate to make it accessible to make it available and we'll be able to organize in a new way and be able to help each other in ways that we wouldn't have expected before so i look forward to it yeah me too great times see you again bye bye-bye [Music] nice awesome stream thanks anna